

# Security Linter Report

|                     |                                |
|---------------------|--------------------------------|
| Filename            | case6_augassign_conditional.py |
| Timestamp           | 2026-07-23 20:06:41 UTC        |
| Lines of Code (LOC) | 4                              |
| Verdict             | VULNERABLE                     |

| ID         | Severity | Sink       | Source Variable | Source Type | Line |
|------------|----------|------------|-----------------|-------------|------|
| CVE-SL-001 | CRITICAL | db.execute | user_filter     | EXTERNAL    | 4    |

|          |   |
|----------|---|
| CRITICAL | 1 |
| WARNING  | 0 |
| INFO     | 0 |
| TOTAL    | 1 |

**CVE-SL-001 — db.execute****CRITICAL**

|                              |                        |
|------------------------------|------------------------|
| <b>CWE</b>                   | CWE-89                 |
| <b>Sink</b>                  | db.execute             |
| <b>Sink Line</b>             | 4                      |
| <b>Sink Column</b>           | 0                      |
| <b>Sink Expression</b>       | db.execute(base_query) |
| <b>Argument</b>              | base_query             |
| <b>Source Variable</b>       | user_filter            |
| <b>Source Type</b>           | EXTERNAL               |
| <b>Propagation Mechanism</b> | data flow propagation  |

***Taint Path Trace:***

| Step | Role   | Line | Expression                                     |
|------|--------|------|------------------------------------------------|
| 1    | SOURCE | 2    | if user_filter:                                |
| 2    | ASSIGN | 3    | base_query += " AND user = " + user_input + "" |
| 3    | SINK   | 4    | db.execute(base_query)                         |

***Remediation:***

Identify the data source and use a parameterized query. Never concatenate external data into SQL strings:

```
# Replace string building with a parameterized query:  
cursor.execute("SELECT * FROM t WHERE col = %s", (external_value,))
```

No safe sink paths confirmed.

No sanitization functions detected.

|                                          |            |
|------------------------------------------|------------|
| CFG — Total Nodes                        | 7          |
| CFG — Condition/Loop Nodes               | 1          |
| CFG — Assignment Nodes                   | 2          |
| CFG — Call Nodes                         | 1          |
| CFG — Merge Nodes                        | 1          |
| ████████████████████████████████████████ | ██████████ |
| DFG — Total Nodes                        | 13         |
| DFG — Total Edges                        | 11         |
| DFG — Source Nodes                       | 0          |
| DFG — Sink Nodes                         | 1          |
| DFG — Operator Nodes                     | 3          |
| DFG — Constant Nodes                     | 3          |
| DFG — Variable Nodes                     | 6          |
| Taint Propagation Hops                   | 3          |
| ████████████████████████████████████████ | ██████████ |
| Analysis Time (ms)                       | 0.1        |

|                        |                                 |
|------------------------|---------------------------------|
| <b>Vulnerabilities</b> | 1 CRITICAL   0 WARNING   0 INFO |
| <b>Safe Sinks</b>      | 0                               |
| <b>Analysis Time</b>   | 0.1 ms                          |
| <b>Engine Version</b>  | 1.0.0                           |
| <b>Final Verdict</b>   | <b>VULNERABLE</b>               |